PROJECT 03: Hospital Billing and Patient Classification

1. Problem Statement

Prepare patient bills from consultation, diagnostic and treatment charges, apply conditional concessions, validate payments and classify payment status.

2. Objectives

• Calculate gross charges from multiple services.
• Apply a 10% concession to eligible senior patients.
• Validate paid amounts and calculate outstanding balance.
• Classify accounts as paid, partially paid, unpaid or invalid.

3. Assumptions

• Patients aged 60 or above receive a 10% concession on the total bill.
• All charge amounts must be non-negative.
• Payment cannot be negative; payments above the bill are treated as paid in full and no cash change is calculated.
• This is a sample billing policy, not a real hospital tariff.

4. Algorithm

1. Read patient age, service charges and amount paid.
2. Validate all monetary inputs.
3. Sum consultation, diagnostic and treatment charges.
4. Apply senior concession if eligible.
5. Calculate payable amount and outstanding balance.
6. Classify payment status and print the statement.

5. Flowchart

START
↓
Input data and validate
↓
Process rules and calculations
↓
Check conditions / classify results
↓
Display result and END

6. Pseudocode

```
START
  FOR each patient
    VALIDATE age, charges and payment
    IF invalid: DISPLAY invalid record
    ELSE
      gross = consultation + diagnostic + treatment
      IF age >= 60: concession = 10% of gross
      ELSE: concession = 0
      payable = gross - concession
      balance = max(0, payable - paid)
      CLASSIFY as paid / partially paid / unpaid
      DISPLAY bill and status
END
```

7. Python Code

```
patients = [
    {"name": "Ravi", "age": 68, "consultation": 500, "diagnostic": 1200, "treatment": 2500,
"paid": 2500},
    {"name": "Meena", "age": 35, "consultation": 500, "diagnostic": 800, "treatment": 1000,
"paid": 2300},
    {"name": "Arun", "age": 42, "consultation": 300, "diagnostic": 0, "treatment": 700, "paid":
```

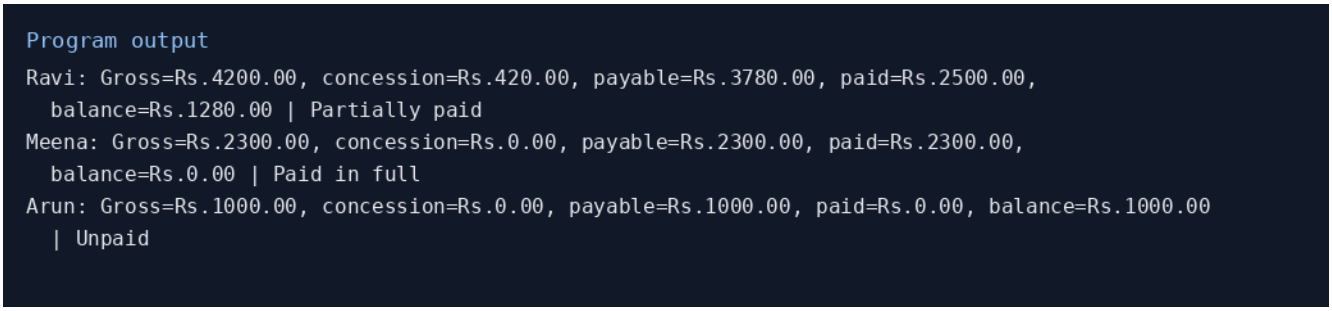
```
0},
]
for p in patients:
    amounts = [p["consultation"], p["diagnostic"], p["treatment"], p["paid"]]
    if p["age"] < 0 or any(v < 0 for v in amounts):
        print(f"{p['name']}: Invalid billing data")
        continue
    gross = sum(amounts[:3])
    concession = gross * 0.10 if p["age"] >= 60 else 0
    payable = gross - concession
    balance = max(0, payable - p["paid"])
    status = "Paid in full" if p["paid"] >= payable else ("Partially paid" if p["paid"] > 0
else "Unpaid")
    print(f"{p['name']}: Gross=Rs.{gross:.2f}, concession=Rs.{concession:.2f}, payable=Rs.{
payable:.2f}, paid=Rs.{p['paid']:.2f}, balance=Rs.{balance:.2f} | {status}")
```

8. Test Cases

Test	Input / condition	Expected result
Senior patient	Age 68, gross charges Rs.4200	10% concession = Rs.420; payable Rs.3780
Adult patient	Age 35, gross Rs.2300	No concession; payable Rs.2300
Unpaid account	Paid amount is zero	Status is Unpaid
Invalid charge	Any charge is negative	Record is rejected as invalid

9. Output Screenshot

The following screenshot-style image was generated from the captured output of the program execution.



Captured output

```
Ravi: Gross=Rs.4200.00, concession=Rs.420.00, payable=Rs.3780.00, paid=Rs.2500.00, balance=Rs.
1280.00 | Partially paid
Meena: Gross=Rs.2300.00, concession=Rs.0.00, payable=Rs.2300.00, paid=Rs.2300.00, balance=Rs.0.
00 | Paid in full
Arun: Gross=Rs.1000.00, concession=Rs.0.00, payable=Rs.1000.00, paid=Rs.0.00, balance=Rs.1000.
00 | Unpaid
```

10. Conclusion

The program provides consistent patient invoices, conditional concessions and clear payment-status classification.

Academic note: The program was executed with the sample data shown in the source code. Policies, rates, thresholds and exchange rates are illustrative and should be adjusted to the assignment's official requirements.